Fixed geometric coordinates, including nullspace | z = 0 mm

Independent phase unwrap; one constant branch shift per curve. Same-record development check.

Same-record held-out development FRF — Additive SRIVC baseline — Fixed-pole structure + channel-delay refinement

$R_x \to R_x$ | $R_y \to R_x$ | $Z \to R_x$ | $N \to R_x$

$R_x \to R_y$ | $R_y \to R_y$ | $Z \to R_y$ | $N \to R_y$

$R_x \to Z$ | $R_y \to Z$ | $Z \to Z$ | $N \to Z$

$R_x \to N$ | $R_y \to N$ | $Z \to N$ | $N \to N$

Unwrapped phase (deg)

Frequency (Hz)